Proof of the chain rule

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$$C(s_1) \leq C(s_2) + C(s_1 \mid s_2) + O(1).$$

Suppose we can find the shortest program p_2 that generates s_2 on our machine. Suppose also that we can find the shortest program p_{21} that generates s_2 from s_1. The length of p_2 is C(s_2), and the length of p_{21} is $C(s_1 \mid s_2)$. The concatenation of p_2 and p_{21} is a program that generates s_1, though not necessarily the shortest one (hence the inequality in the chain rule).

This only holds for "unambiguous" program codes (so that the machine knows where p_2 ends and where $p_2 1$ begins).

Close